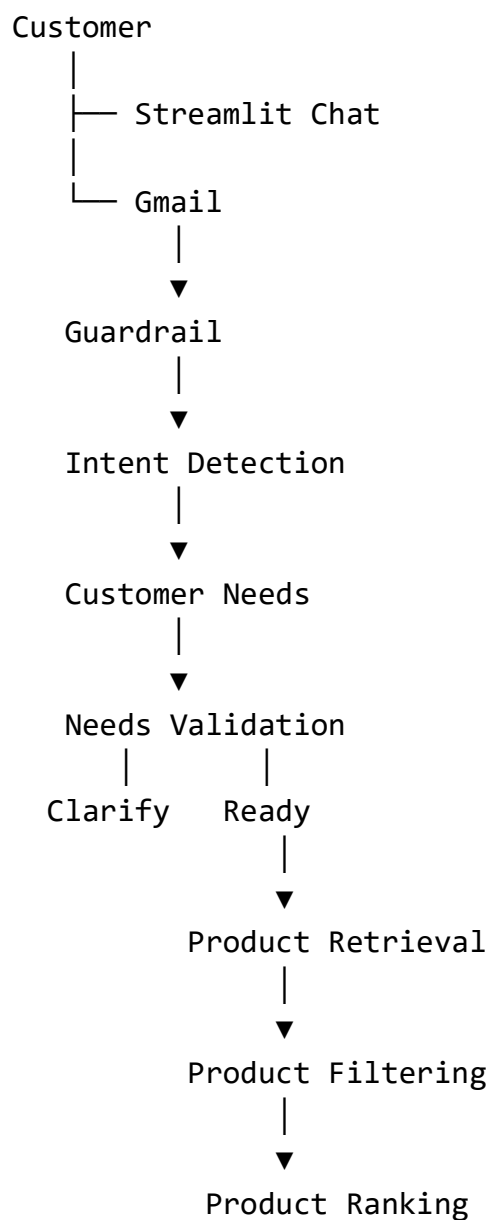


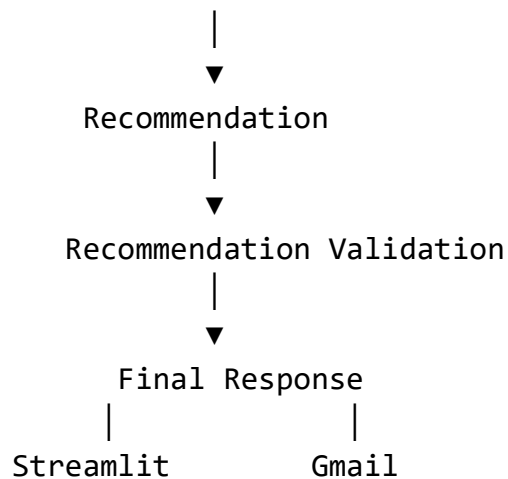
Oak & Ember Interiors — AI Sales Agent

1. Project Overview

An end-to-end AI furniture sales assistant for **Oak & Ember Interiors**. It accepts customer queries through **Streamlit chat or Gmail**, understands customer requirements, retrieves and ranks relevant products, validates recommendations, and returns a formatted response.

2. Architecture





3. Technology Stack

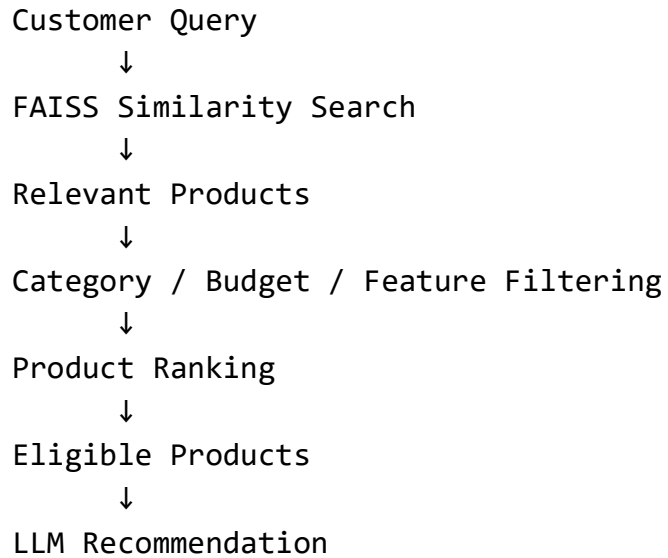
- **Python**
- **LangGraph** – agent workflow orchestration
- **LangChain** – LLM and tool integration
- **OpenAI** – LLM
- **FAISS** – vector database
- **Pandas** – product filtering and processing
- **Pydantic** – structured data validation
- **Streamlit** – chat UI
- **Gmail API** – email integration
- **Langfuse** – observability and tracing
- **LangSmith / Evaluation framework** – evaluation support

4. Agent Workflow

The LangGraph workflow contains:

- Input guardrail
- Intent classification
- Customer needs extraction
- Needs validation and clarification
- Product retrieval
- Product filtering
- Product ranking
- Recommendation generation
- Recommendation validation
- Final response generation

5. RAG Pipeline



6. Guardrails

The input guardrail uses:

- Regex-based prompt-injection detection
- LLM-based classification
- Categories: safe, prompt_injection, unsafe, off_topic

Blocked requests are routed to a dedicated response node.

7. Evaluation

A **golden dataset of customer queries** is used to evaluate the agent.

Evaluation metrics include:

- Correctness
- Relevance
- Hallucination
- Groundedness

Individual evaluation scores are recorded for each golden-dataset query and aggregated for overall evaluation results.

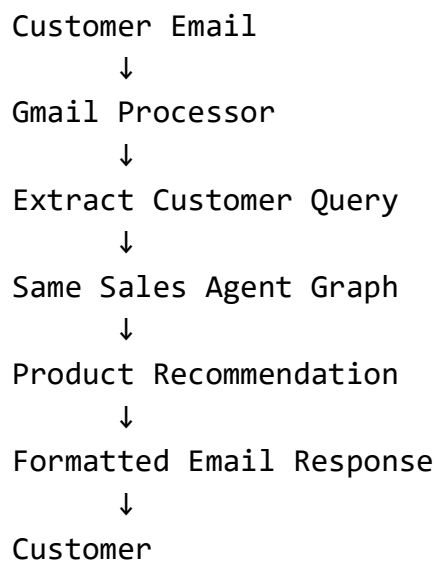
8. Observability

Langfuse is used to monitor the application, including:

- Agent workflow traces
- Important workflow spans
- LLM generations
- Guardrail results
- Retrieval information
- Evaluation scores
- Latency
- Token usage
- Model/cost information

9. Gmail Integration

The Gmail flow:



No human approval step is used.

10. Running the Application

Install dependencies

```
pip install -r requirements.txt
```

Configure environment variables

Create a `.env` file containing the required OpenAI and Langfuse credentials.

Start Streamlit

```
Streamlit start_app.py
```

Start Gmail processing

```
python gmail_processor.py
```

The Streamlit interface handles chat-based queries, while the Gmail processor handles incoming customer emails.